

Wooseong Jeong

Ph.D Candidate, KAIST
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EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** Daejeon, Korea
Ph.D. Candidate in Mechanical Engineering March 2023 – August 2027 (Expected)
Advisor: Kuk-Jin Yoon
- **Korea Advanced Institute of Science and Technology (KAIST)** Daejeon, Korea
M.S. in Mechanical Engineering March 2021 – February 2023
Advisor: Kuk-Jin Yoon
Thesis: Connection Strength-based Optimization with Progressive Multi-modal Feature Exchange for Multi-task Learning
- **Korea Advanced Institute of Science and Technology (KAIST)** Daejeon, Korea
B.S. in Mechanical Engineering March 2015 – February 2021
Double Major: Mechanical Engineering and Electrical Engineering, cum laude
Dean's List: Fall 2017

RESEARCH INTERESTS

- Parameter-efficient learning and model merging (LoRA/PEFT, conflict-aware weight composition, preference-aligned merging)
- Multi-task learning and scalable optimization (task interaction, Pareto optimization, preference-based multi-objective learning)
- Test-time adaptation and robust learning under distribution shift
- Autonomous driving and robotics (motion planning, vision-language-action models, robust autonomy)

PUBLICATIONS

- **Wooseong Jeong***, Wonyoung Lee*, and Kuk-Jin Yoon, "Preference-Aligned LoRA Merging: Preserving Subspace Coverage and Addressing Directional Anisotropy", **CVPR 2026**.
- Wonyoung Lee*, **Wooseong Jeong***, and Kuk-Jin Yoon, "Label-Free Cross-Task LoRA Merging with Null-Space Compression", **CVPR 2026**.
- **Wooseong Jeong***, Jegyeong Cho*, Youngho Yoon*, Jaeyoung Lee, and Kuk-Jin Yoon, "Stabilizing Multi-Task Latent Spaces: Recursive Refinement with Coordinators in Partially Labeled Learning", **IEEE Access 2026**.
- Giwon Lee*, **Wooseong Jeong***, Daehee Park, Jaewoo Jeong, and Kuk-Jin Yoon, "Interaction-Merged Motion Planning: Effectively Leveraging Diverse Motion Datasets for Robust Planning", **ICCV 2025 (Highlight)**.
- **Wooseong Jeong** and Kuk-Jin Yoon, "Resolving Token-Space Gradient Conflicts: Token Space Manipulation for Transformer-Based Multi-Task Learning", **ICCV 2025**.
- **Wooseong Jeong***, Jegyeong Cho*, Youngho Yoon*, and Kuk-Jin Yoon, "Synchronizing Task Behavior: Aligning Multiple Tasks during Test-Time Training", **ICCV 2025**.
- Jihun Kim*, Hoyong Kwon*, Hyeokjun Kweon*, **Wooseong Jeong**, and Kuk-Jin Yoon, "DC-TTA: Divide-and-Conquer Framework for Test-Time Adaptation of Interactive Segmentation", **ICCV 2025**.
- **Wooseong Jeong***, Jihun Kim*, Hyeokjun Kweon*, and Kuk-Jin Yoon, "Multi-View 3D Scene Abstraction From Drone-Captured RGB Images", **IEEE Access 2025**.
- **Wooseong Jeong** and Kuk-Jin Yoon, "Selective Task Group Updates for Multi-Task Optimization", **ICLR 2025**.
- **Wooseong Jeong** and Kuk-Jin Yoon, "Quantifying Task Priority for Multi-Task Optimization", **CVPR 2024**.
- Hyeokjun Kweon*, Hyeonseong Kim*, Yoonsu Kang*, Youngho Yoon*, **Wooseong Jeong**, and Kuk-Jin Yoon, "Pixel-wise Warping for Deep Image Stitching", **AAAI 2023**.

MANUSCRIPTS UNDER REVIEW

- **Wooseong Jeong***, Giwon Lee*, and Kuk-Jin Yoon, "Latent Reasoning Path Guidance: Adapting ICL Samplers for Cross-Domain Retrieval", **EMNLP 2026** (Under Review).
- Giwon Lee*, **Wooseong Jeong***, and Kuk-Jin Yoon, "CHORUS: Adapting the Number of Demonstrations via Test-Time LLM Consensus", target: **NeurIPS 2026** (Under Review).

- Hyemin Yang*, **Wooseong Jeong***, Giwon Lee*, and Kuk-Jin Yoon, “Triggering Generalist Reasoning via Predictive Uncertainty for Dual-System VLA”, target: **NeurIPS 2026** (Under Review).
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- IN PREPARATION
- **Wooseong Jeong**, Giwon Lee, and Kuk-Jin Yoon, “Cross-Scenario Adaptation via Modular Knowledge Merging in E2E Autonomous Driving”, target: **RA-L**.
 - Yujeong Chae*, **Wooseong Jeong***, and Kuk-Jin Yoon, “LiDAR-RADAR Rebalancing for 3D Detection under Domain Shifts and Sensor Failure”, **AAAI 2026**.

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- PATENTS
- Kuk-Jin Yoon, Hyeokjun Kweon, Yoonsu Kang, **Wooseong Jeong**, Hyeonseong Kim, and Hwi-Sung Park, “Apparatus and Method for Generating Image Dataset for Learning and Evaluation of Image Stitching”, Republic of Korea Patent No. 10-2645640, Registered Mar. 5, 2024.
 - Kuk-Jin Yoon, Hyeokjun Kweon, Yoonsu Kang, **Wooseong Jeong**, Hyeonseong Kim, and Hwi-Sung Park, “Apparatus and Method for Image Stitching”, Republic of Korea Patent No. 10-2571530, Registered Aug. 23, 2023.

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- RESEARCH EXPERIENCE
- **Project Leader** March 2025 – Present
Core Technology Development for Autonomous Driving in Unstructured Off-Road Environments
Funded by Hanwha Aerospace. Real-world deployment target in unstructured off-road terrain.
 - **Project Leader** March 2024 – March 2025
Unmanned Swarm CPS Research Lab
Funded by Agency for Defense Development (ADD). Multi-agent perception and coordination for autonomous robotic systems.
 - **Project Leader** March 2022 – February 2024
Development of a 2kg-Class Small Electro-Optical Tracking System for Drones
Funded by Institute of Civil-Military Technology Cooperation. Multi-modal (EO/IR) detection and tracking deployed on drone hardware.
 - **Project Member** March 2021 – February 2022
Unmanned Swarm CPS Research Lab
Funded by Agency for Defense Development (ADD). Multi-agent perception research for autonomous robotic systems.

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- ACADEMIC SERVICE
- Reviewer for NeurIPS (2024, 2025), ICLR (2025, 2026), CVPR (2025, 2026), ICML (2025), ICCV (2025), ECCV (2026).

SKILLS	
Languages	Python (proficient), C++ (proficient), MATLAB
ML Frameworks	PyTorch, JAX, NumPy, OpenCV
Efficient Training	PyTorch DDP, DeepSpeed, flash-attention, xformers
Hugging Face	Transformers, Diffusers, PEFT, Accelerate
Experiment & DevOps	Weights & Biases, TensorBoard, Docker, Git
Robotics	ROS, ROS2
Spoken Languages	Korean (native), English (fluent)